

-- ===== -- PERSONAL EXPENSE TRACKER DATABASE -- SQL
PRACTICAL FILE -- =====

-- TABLE STRUCTURE (GIVEN - DO NOT MODIFY)

CREATE TABLE users (user_id INT PRIMARY KEY, name VARCHAR(50), email VARCHAR(100), created_at
DATE);

CREATE TABLE categories (category_id INT PRIMARY KEY, category_name VARCHAR(50));

CREATE TABLE expenses (expense_id INT PRIMARY KEY, user_id INT, category_id INT, amount
DECIMAL(10,2), expense_date DATE, FOREIGN KEY (user_id) REFERENCES users(user_id), FOREIGN KEY
(category_id) REFERENCES categories(category_id));

-- INSERT USERS

INSERT INTO users VALUES (1, 'Amit Sharma', 'amit@gmail.com', '2026-05-01'), (2, 'Priya Patel',
'priya@gmail.com', '2026-05-02'), (3, 'Rahul Verma', 'rahul@gmail.com', '2026-05-03'), (4, 'Sneha Joshi',
'sneha@gmail.com', '2026-05-04'), (5, 'Karan Mehta', 'karan@gmail.com', '2026-05-05');

-- INSERT CATEGORIES

INSERT INTO categories VALUES (1, 'Food'), (2, 'Rent'), (3, 'Entertainment');

-- INSERT EXPENSES

INSERT INTO expenses VALUES (1, 1, 1, 250.00, '2026-05-10'), (2, 1, 2, 5000.00, '2026-05-11'), (3, 2, 1,
300.00, '2026-05-11'), (4, 2, 3, 800.00, '2026-05-12'), (5, 3, 2, 7000.00, '2026-05-13'), (6, 3, 1, 450.00,
'2026-05-14'), (7, 4, 3, 1200.00, '2026-05-15'), (8, 4, 1, 350.00, '2026-05-16'), (9, 5, 2, 6500.00, '2026-05-17'),
(10, 5, 3, 900.00, '2026-05-18');

-- UPDATE EXPENSE

UPDATE expenses SET amount = 600.00 WHERE expense_id = 4;

-- DELETE EXPENSE

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DELETE FROM expenses WHERE amount < 300;
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-- JOIN QUERY (FULL DETAILS)
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```
SELECT e.expense_date, e.amount, u.name, c.category_name FROM expenses e INNER JOIN users u ON  
e.user_id = u.user_id INNER JOIN categories c ON e.category_id = c.category_id;
```

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-- TOTAL SPENDING PER CATEGORY
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```
SELECT c.category_name, SUM(e.amount) AS total_expense FROM expenses e INNER JOIN categories c  
ON e.category_id = c.category_id GROUP BY c.category_name;
```

```
-- USERS SORTED BY SPENDING
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```
SELECT u.name, SUM(e.amount) AS total_spending FROM expenses e INNER JOIN users u ON e.user_id =  
u.user_id GROUP BY u.name ORDER BY total_spending DESC;
```

```
-- VIEW: ACTIVE USERS (>5 EXPENSES)
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```
CREATE VIEW ActiveUsersView AS SELECT u.name, u.email FROM users u JOIN expenses e ON u.user_id  
= e.user_id GROUP BY u.user_id, u.name, u.email HAVING COUNT(e.expense_id) > 5;
```

```
-- VIEW QUERY
```

```
SELECT * FROM ActiveUsersView;
```

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-- STORED PROCEDURE -- MONTHLY EXPENSE CALCULATION
```

```
DELIMITER //
```

```
CREATE PROCEDURE MonthlyExpense(IN uid INT) BEGIN SELECT user_id, MONTH(expense_date) AS  
month, SUM(amount) AS total_monthly_expense FROM expenses WHERE user_id = uid GROUP BY  
MONTH(expense_date); END //
```

DELIMITER ;

-- PROCEDURE CALL

CALL MonthlyExpense(1);

-- TRANSACTION EXAMPLE -- COMMIT

START TRANSACTION;

INSERT INTO expenses VALUES (11, 2, 1, 400.00, '2026-05-21');

COMMIT;

-- TRANSACTION EXAMPLE -- ROLLBACK

START TRANSACTION;

INSERT INTO expenses VALUES (12, 3, 2, 999.00, '2026-05-22');

ROLLBACK;

-- ===== -- END OF FILE --
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